

Vatsal Yadav

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Professional Summary

Third-year B.Tech student in Data Science & AI at IIT Bhilai targeting AI/ML, LLM, and software engineering intern roles. Built RAG pipelines, multi-agent systems, multimodal AI, RESTful backends, and ML models, with strong fundamentals in data structures, algorithms, OOP, and system design. Production internship experience with FastAPI, LangGraph, Docker, and AWS.

Education

Indian Institute of Technology (IIT) Bhilai

2024 – 2028

Bachelor of Technology in Data Science & Artificial Intelligence

Bhilai, Chhattisgarh

Coursework: Data Structures, Algorithms, Computer Organisation & Architecture, Statistical Programming, Analytics

Experience

Incrivelssoft Private Limited (NUMAA.ai)

May 2026 – July 2026

AI & ML Intern

Remote

- Architected the Nutritionist Lite Agent for the NUMAA.ai multi-agent platform as a 3-layer hybrid: deterministic clinical engine (ICMR-NIN 2020 / WHO), guideline retrieval, and a Google Gemini (gemini-2.5-flash) router; enforced hard medical-nutrition boundaries to curb hallucinations.
- Engineered 5 backend services with FastAPI, Pydantic, and MongoDB; built a Qdrant RAG pipeline (ChromaDB prototype) and a Gemini Vision food-photo endpoint with confidence-based fallback.
- Led migration from 3 LLM providers (Groq, Google, NVIDIA) to Gemini; added token-bucket rate limiting (300 RPM), a circuit breaker (5 failures / 30s cooldown), and model fallback, eliminating HTTP 429 errors.
- Refactored NutritionUIService into modular caching, repository, and prompt layers using OOP (~855 to ~480 lines); wrote 31 unit tests with a FakeProfileRepository adapter for MongoDB-free testing.

Projects

Context Pager | Python, MCP, LLMingua-2, BGE-M3, FastAPI, Docker, AWS

GitHub | Live

- Built an MCP (Model Context Protocol) document intelligence server cutting AI-agent token usage by 4–10x via compressed paging; exposes search, compression, and memory tools to Claude Desktop, Claude Code, and Cursor.
- Designed a privacy-first three-tier system (local Bridge, cloud Relay, Agent) on AWS t3.micro with PII masking (Microsoft Presidio), SHA-256 hashed API keys, and rate limiting at 100 calls/hour.

HelixDesk – Enterprise Support Intelligence | Python, LangGraph, FastAPI, Pinecone, Docker

GitHub | Live

- Shipped a production-grade RESTful API backend (async FastAPI + Pydantic) driving a 3-agent LangGraph state machine (Triage, Retrieval, Resolution) on Llama-3.3-70b; escalates tickets below 50% confidence to humans.
- Built Pinecone semantic search with multilingual-e5-large embeddings and citation-backed answers, structured JSON logging, X-Correlation-ID request tracing, and GitHub Actions CI/CD; achieved 1840 ms average resolution.

JobFit-AI – Resume Matching Engine | Python, XGBoost, PyTorch, SBERT, Docker, AWS

GitHub | Live

- Engineered a 3-model resume-JD matcher: rule-based skill scorer (430+ skills, 24 categories), XGBoost on 10 features, and a fine-tuned SBERT dual-encoder (all-MiniLM-L6-v2) using BatchNorm, GELU, dropout, OneCycleLR, and early stopping; trained on 13,000+ resume-JD pairs.
- Applied rank-based label scoring (mean 50, std 21.5) for balanced calibration; containerized with Docker (CPU-only PyTorch) and deployed on AWS EC2 (t3.small) with auto-restart and a Streamlit inference UI.

Technical Skills

Languages: Python, C, JavaScript, SQL

Core CS: Data Structures & Algorithms, System Design, Object-Oriented Programming (OOP), Unit Testing, Git

AI / LLMs: LangChain, LangGraph, Multi-Agent Systems, RAG, MCP, Prompt Engineering, Fine-tuning (LoRA, PEFT), Multimodal AI, Vision-Language Models, Embedding Models, Google Gemini API, OpenAI API, Hugging Face, Vector Databases (Pinecone, Qdrant, ChromaDB)

ML / Deep Learning: PyTorch, Scikit-learn, XGBoost, Transformers, Neural Networks, Sentence-Transformers, NLP, NumPy, Pandas, EDA, Feature Engineering

Backend & Cloud: FastAPI, RESTful APIs, Docker, AWS (EC2, ECR), GitHub Actions (CI/CD), MongoDB

Leadership & Achievements

Data Science and AI Club (DSAI), IIT Bhilai | Coordinator & Core Member

Aug 2024 – Present

- Maintain an open-source AI/ML Compendium; led Agentic AI sessions; organized 100+ participant Meraz hackathon.
- Campus Finalist, The Integral Cup (2024): quantitative reasoning contest by Optiver, QRT, Jane Street.